Rajalakshmi Engineering College

Department of Computer Science and Engineering

LAB RECORD

Library Book Issue Automation Suite - Workflow Options

Detail	
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Course / Section	B E CSE
Platform	Zoho Creator
Application	Library Book Issue Management
Assignment	Assignment 06 - Workflow Options

1. Aim

To design and build an automation layer on the Book_Issue_Request_Form in Zoho Creator using all eight Workflow options (Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions), so that confirmations, reminders, approvals, fine collection, overdue notices and gate-pass generation run with minimal manual effort.

2. Procedure

2.1 Preparing the application

1. Opened the application Library Book Issue Management in Zoho Creator and used Book_Issue_Request_Form and Book_Issue_Request_Report as the base form and report.
2. Confirmed that the Status, Due Date, Fine_Amount_Due, Late_Fee_Waiver_Percent, Total_Fine_Payable and Fine_Paid_Status fields are available, and added sample records.

2.2 Workflow options used

Option	Workflow name	Purpose	Trigger / frequency
Form Workflow	On_IssueRequest_Submit_Workflow	Confirmation email with Issue Request ID to the member; notifies counter staff	Instantly on Add of Book_Issue_Request_Form
Schedule	Daily_DueDate_Reminder_Schedule	Reminder email for books due within the next 2 days	Daily at 8:00 AM
Approval	RareBook_Issue_Approval	Routes rare/reference-only book requests to the librarian for approval	On Add/Edit when Book Category is Rare/Reference-Only

Option	Workflow name	Purpose	Trigger / frequency
Payments	Fine_Payment_Workflow	Payment link for outstanding fine; updates Fine_Paid_Status	When a member with Fine_Amount_Due > 0 tries to issue a new book
Blueprint	Issue_Lifecycle_Blueprint	Controlled lifecycle with role-based transitions	Every record; Requested > Approved > Issued > Returned / Overdue
Batch Workflow	Bulk_Overdue_Notice_Batch	Overdue notice with current fine, processed in batches	Nightly at 10:00 PM; batches of 200
Report Workflow	OnEdit_Recalculate_Fine_Workflow	Recalculates Total_Fine_Payable when a waiver is edited in the report	On Edit of any record in Book_Issue_Request_Report
Function	Generate_GatePass_Function	Generates a gate-pass/receipt PDF and emails it to the member	When Status changes to Issued; also via Reprint Gate Pass button

2.3 Building and configuring the workflows

1. Form Workflow: created On_IssueRequest_Submit_Workflow on Add of Book_Issue_Request_Form to email the member their Issue Request ID and expected collection time and to notify counter staff. Set it to log an error and continue if the member's email is invalid, instead of blocking the request.
2. Schedule: created Daily_DueDate_Reminder_Schedule to run every day at 8:00 AM, pick records whose Due Date falls within the next 2 days, and email those members. Set it to send only one reminder per book and to stop once the book is marked Returned.
3. Approval: created RareBook_Issue_Approval for requests whose Book Category is Rare/Reference-Only, keeping Status at Pending Approval until the librarian approves or rejects. Set the Librarian as approver and the Assistant Librarian as fallback; regular books bypass this step.
4. Payments: configured Fine_Payment_Workflow with a payment gateway connector to generate a payment link from Total_Fine_Payable and update Fine_Paid_Status to Paid on success, or keep the account restricted on failure. Set the gateway status-check callback to retry up to 3 times before flagging the transaction as failed.
5. Blueprint: created Issue_Lifecycle_Blueprint with the stages Requested, Approved, Issued, Returned and Overdue, restricted each transition to specific roles, and made a comment mandatory on every transition for an audit trail.
6. Batch Workflow: created Bulk_Overdue_Notice_Batch to run every night at 10:00 PM, find members with overdue books, and email a consolidated notice listing pending titles and the current fine. Set the batch size to 200 records.
7. Report Workflow: created OnEdit_Recalculate_Fine_Workflow on Book_Issue_Request_Report to recalculate Total_Fine_Payable, firing only when Fine Amount Due or Late Fee Waiver Percent changes.
8. Function: wrote Generate_GatePass_Function once in Deluge to create the gate-pass PDF with the issued titles and due date and email it to the member. Added input validation so it fails safely if Member Name or Book Title is empty, called it when Status changes to Issued, and added a Reprint Gate Pass button for manual calls.

3. Output

(a) Workflow configuration screens

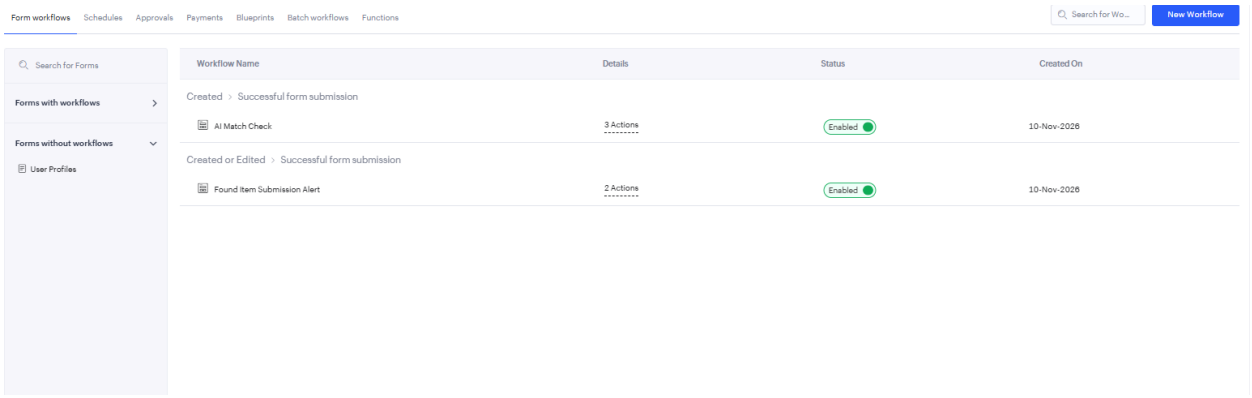
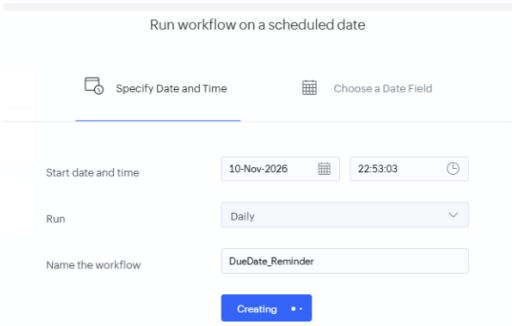


Figure 1: On_IssueRequest_Submit_Workflow (Form Workflow) in the Workflow Builder



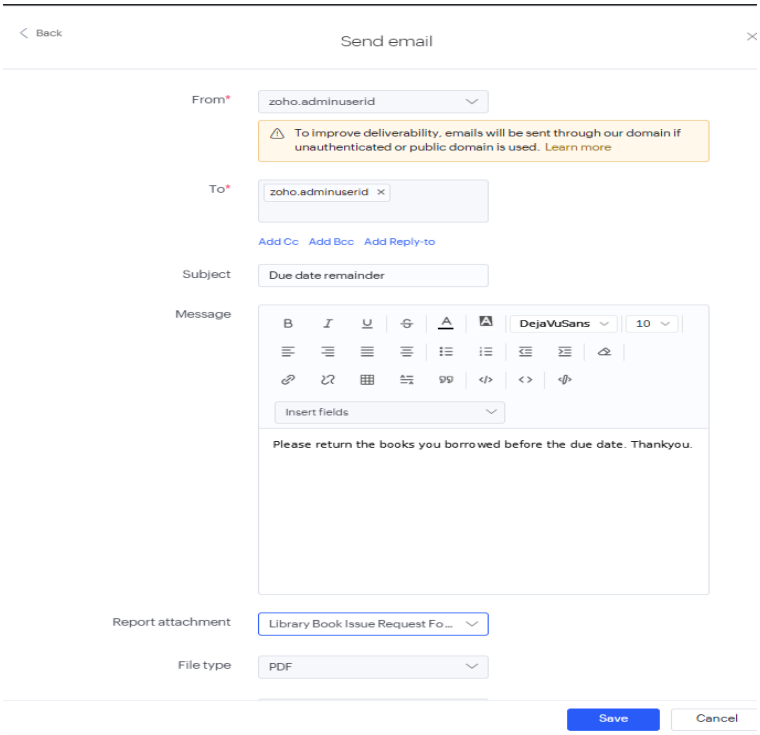


Figure 2: Daily_DueDate_Reminder_Schedule (Schedule) configuration

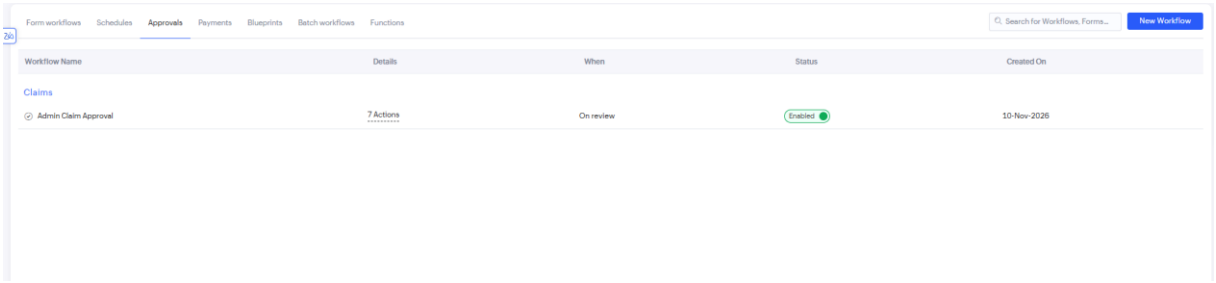


Figure 3: RareBook_Issue_Approval (Approval) configuration

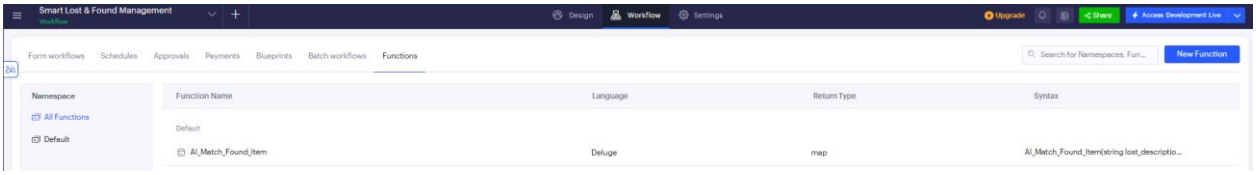


Figure 4: Generate_GatePass_Function (Function) configuration

(b) End-to-end sample run

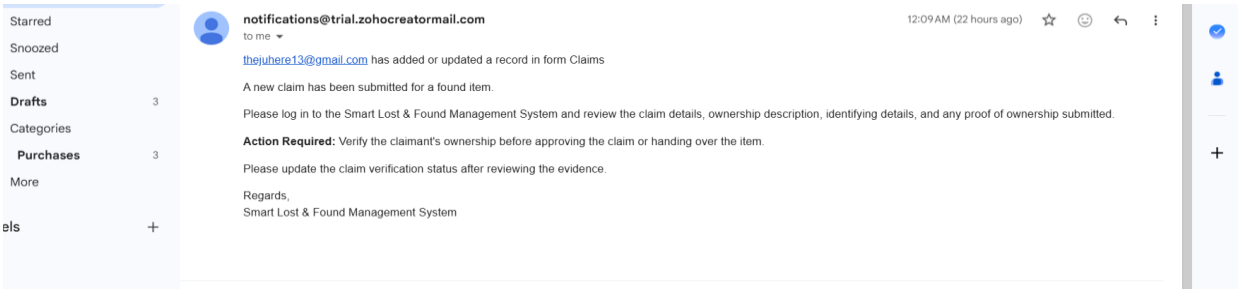


Figure 5: Sample confirmation email received by the member

(c) Generate_GatePass_Function

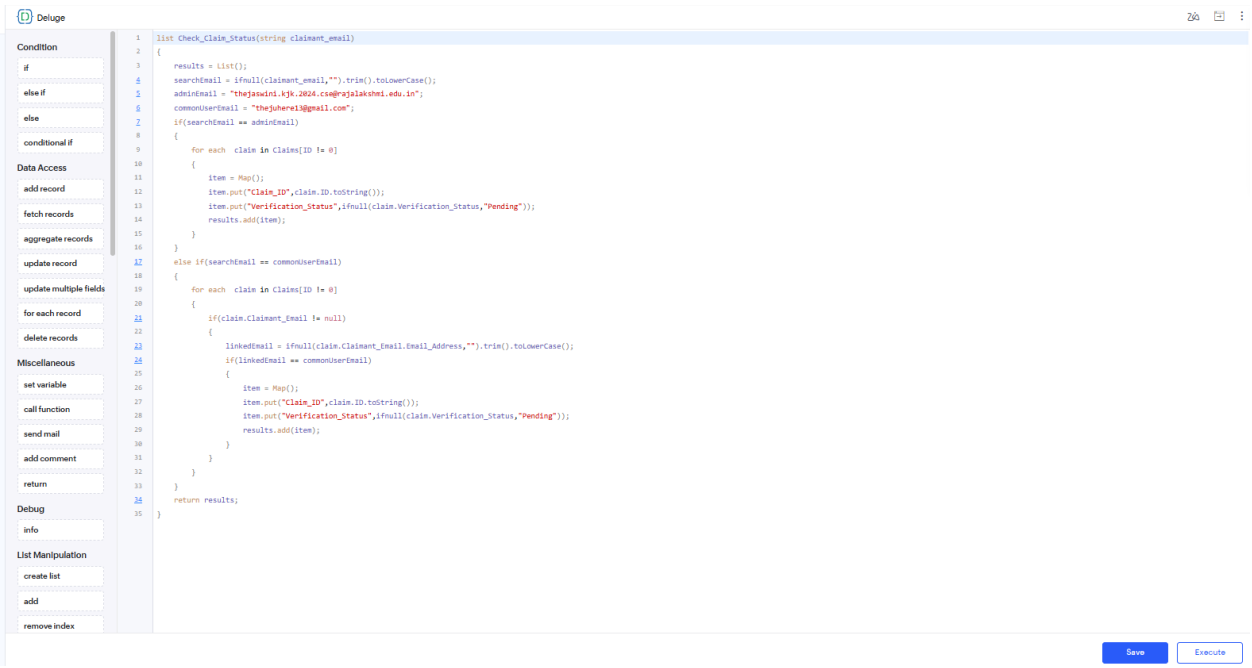


Figure 6: Deluge script of Generate_GatePass_Function

Result:

All eight workflow options (Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions) were built on the Book_Issue_Request_Form in Zoho Creator and the governance rules were applied. The confirmation email, fine-payment link, Blueprint lifecycle and gate-pass function worked end to end on the sample records.